# Appendix 1 – Palettes by category

**Important.**

*The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.*

**Note** (Qualitative and semantic palettes).

*The threshold of 40 colors has been established for qualitative and semantic palettes. The following lines outline the rationale for this limit.*

- *In qualitative color palettes, each color represents a single data point or a group of data. Experience shows that about ten colors are more than enough.*
- *Semantic palettes associate a color with a specific context, such as warning levels (see the three frames in this document). Choosing 40 colors allows for 10 different contexts, each with 4 sub-categories. Here are some examples of sub-categories.*

  ***10-19*** ***Initial / Idle:*** *default states such as "To Do", "Clickable", or "To be analyzed".*
  ***20-29*** ***Active / Transition:*** *states in flux, including "In Progress" or UI "Hover" effects.*
  ***30-39*** ***Success / Final:*** *terminal positive states like "Done", "Clicked", or "Validated".*
  ***40-49*** ***Alert / Rejection:*** *error handling and negative outcomes, e.g., "Error", "Inactive", or "Rejected".*

**Important.**

*Categories were generated semi-automatically, followed by manual selection to obtain relevant choices.*

## Colorblind palettes

Colorblind palettes provide high-contrast hues specifically optimized for users with color vision deficiency.

**Acton**

**Apple**

**Bam**

**Bamako**

**BamO**

**Batlow**

**BatlowK**

**BatlowW**

**Berlin**

**Bilbao**

**Broc**

**BrocO**

**Buda**

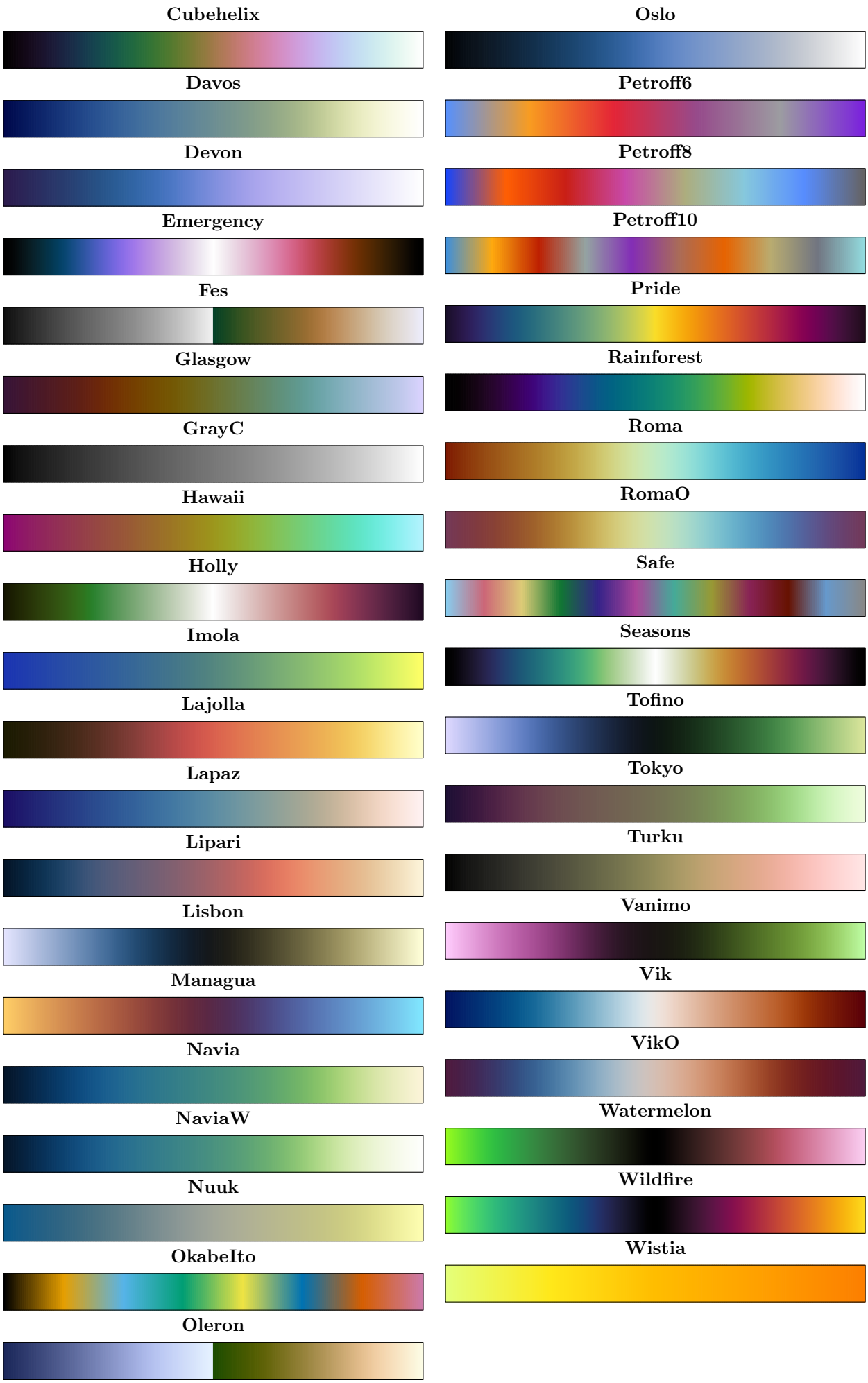
**Bukavu**

**Chroma**

**CopperCM**

**Cork**

**CorkO**


Cubehelix
Davos
Devon
Emergency
Fes
Glasgow
GrayC
Hawaii
Holly
Imola
Lajolla
Lapaz
Lipari
Lisbon
Managua
Navia
NaviaW
Nuuk
OkabeIto
Oleron
Oslo
Petroff6
Petroff8
Petroff10
Pride
Rainforest
Roma
RomaO
Safe
Seasons
Tofino
Tokyo
Turku
Vanimo
Vik
VikO
Watermelon
Wildfire
Wistia

## Cyclic palettes

Cyclic palettes create seamless loops for periodic data, such as angles, seasons, or 24-hour cycles.

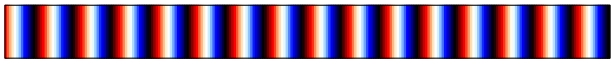
**Balance**

**BamO**

**CopperCM**

**Emergency**

**Flag**



**Infinity**

**RainbowCP**

**Seasons**

**SineBow**

## Dark palettes

Dark palettes offer high-contrast schemes specifically optimized for terminal interfaces and dark-mode backgrounds.

**Autumn**

**BWR**

**Cool**

**GistRainbow**

**HSV**

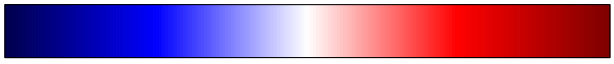
**Jet**

**Lemon**

**Prism**

**Rainbow**

**Seismic**



**Spring**

**Summer**

**Tab20**

**Tab20b**

**Tab20c**

**TdocDark**

**Terrain**

**Viridis**

**Winter**

**YlOrRd**

## Divergent palettes

Divergent palettes highlight deviations from one midpoint toward contrasting extremes.

**ArmyRose**

**Balance**

**Bam**

**BamO**

**Berlin**

**Blackbody**

**BlindFish**

**BlueDarkOrange**

**BlueDarkRed12**

**BlueDarkRed18**

**BlueGray**

**BlueGreen**

**BlueGrey**

**BlueOrange**

**BlueOrangeRed**

**BrBG**

**BrownBlue10**

**BrownBlue12**

**BurningGrass**

**BWR**

**CoolWarm**

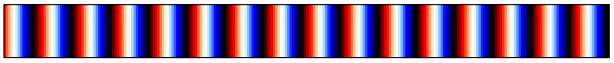
**Curl**

**Delta**

**Earth**

**Fall**

**Flag**



**Fusion**

**Geyser**

**GistNcar**

**GistRainbow**

**GistStern**

**GreenGrey**

**GreenMagenta**

**Guppy**

**Holly**

**HSV**

IceBurn
Jet
NipySpectral
Ocean
OrangeBlue
OrangeGrey
Picnic
PinkGreen
PinkGrey
PiYG
Portland
PrGn
Pride
Prinsenvlag
Prism
PuOr
PurpleGrey
RdBu
RdGy
RdYlBu
RdYlGn
RedGrey
RedShift
RedTeal
RedYellowBlue
Seaweed
Seismic
Spectral
Tab20
Tab20b
Tab20c
TealGrey
TealRose
Temps
Terrain
Tropic
Twilight
Viola
Waterlily
Watermelon

**Wildfire**

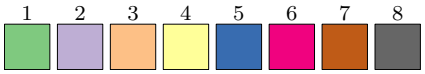
**YellowGrey**

**YellowPurple**

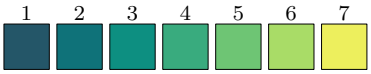
## Qualitative palettes

Qualitative palettes use distinct hues to separate categorical data without implying any order or hierarchy.

**Accent**

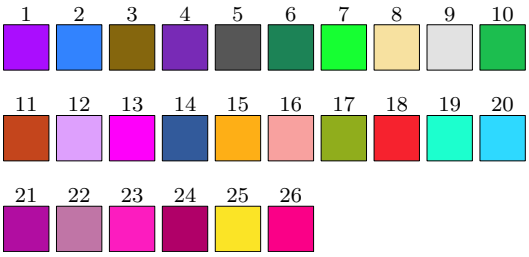



**AgGrnYl**

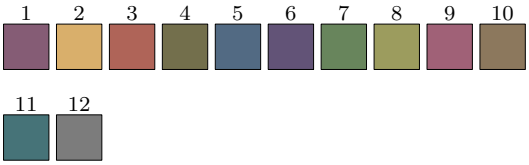



**AgSunset**

**Alphabet**

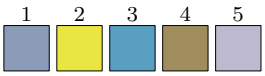



**Antique**

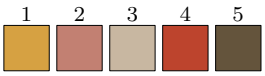



**Aquatic1**

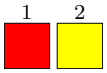
**Aquatic2**



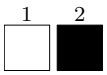

**Aquatic3**



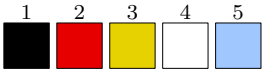

**Autumn**



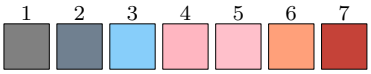

**Binary**



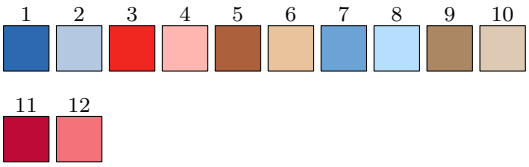

**Blackbody**



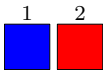

**BlindFish**



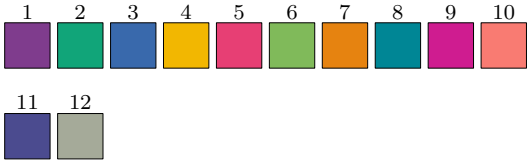

**BlueRed**



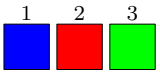

**BlueRedPLY**




**Bold**

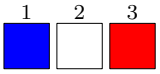



**BRG**

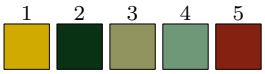



**BurningGrass**

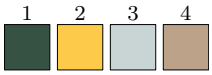
**BWR**



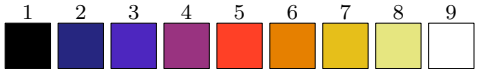

**Cavalcanti**



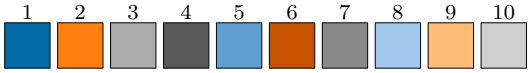

**Chevalier**



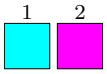

**CMRmap**




**ColorBlind**

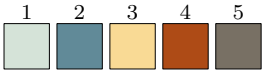



**Cool**

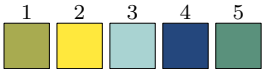



**Darjeeling1**

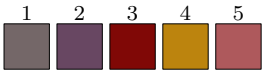
**Darjeeling2**




**Darjeeling3**

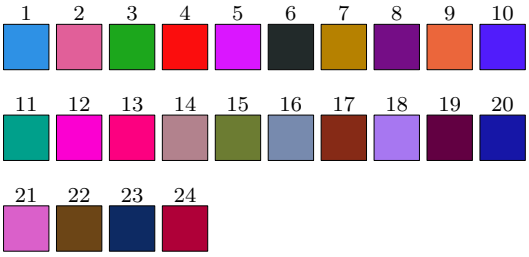



**Darjeeling4**

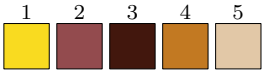



**Dark2**

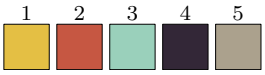
**Dark24**




**FantasticFox1**

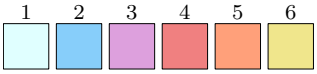



**FantasticFox2**



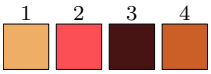

**G10**

**GasFlame**

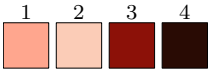



**GeoRainbow**

**GrandBudapest1**

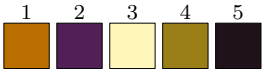



**GrandBudapest2**

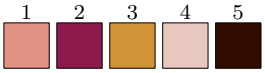



**GrandBudapest3**

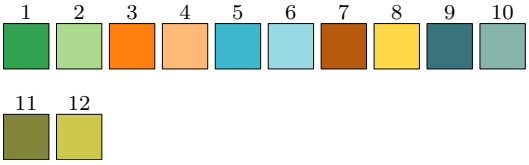
**GrandBudapest4**



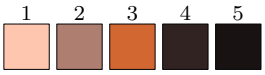

**GrandBudapest5**



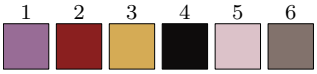

**GreenOrange**



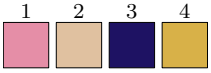

**IsleOfDogs1**



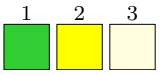

**IsleOfDogs2**



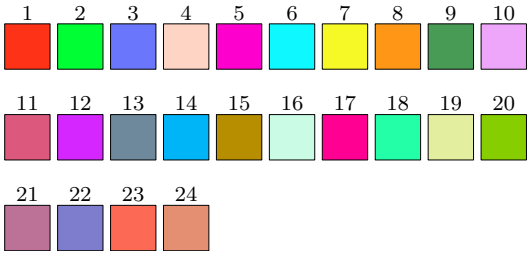

**IsleOfDogs3**




**Lemon**

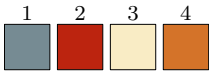



**Light24**

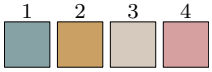



**Margot1**

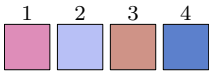
**Margot2**



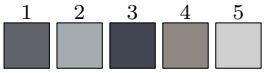

**Margot3**



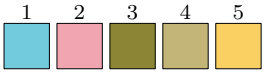

**Mendl**




**MidGray**

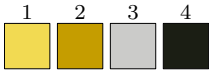



**Moonrise1**

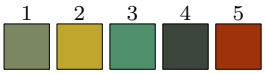



**Moonrise2**

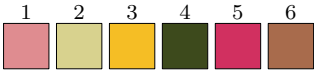
**Moonrise3**



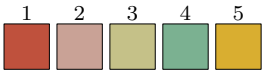

**Moonrise4**



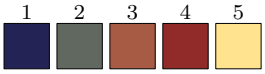

**Moonrise5**



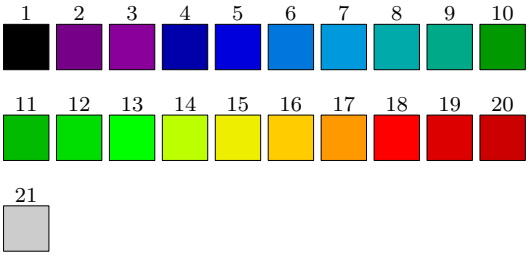

**Moonrise6**



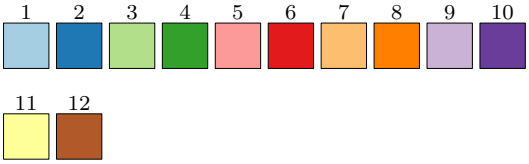

**Moonrise7**



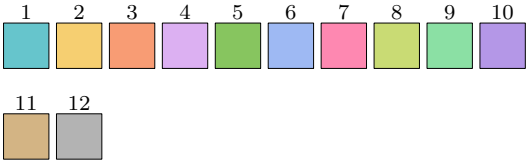
### NipySpectral



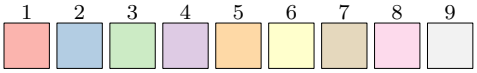

### Paired



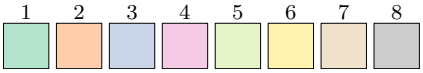

### Pastel



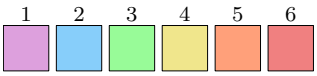

### Pastel1



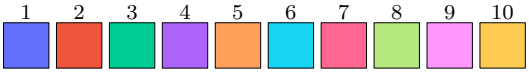

### Pastel2



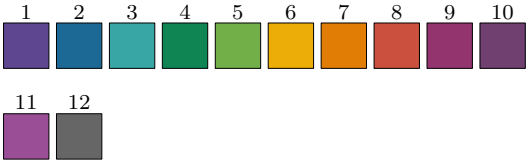

### PastelRainbow



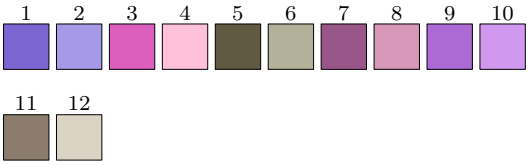

### Plotly




### PrismCC

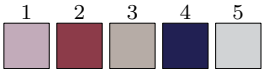



### PurpleGray



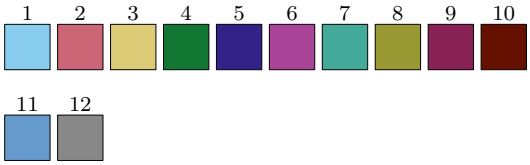

### Royal1

### Royal2

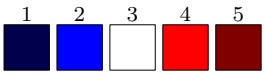



### Royal3

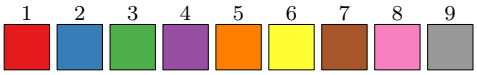
### Safe



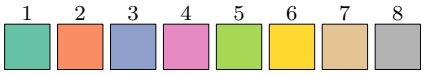

### Seismic



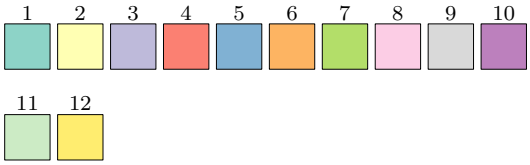

### Set1




### Set2

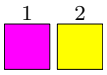



### Set3

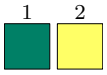



### ShiftRainbow

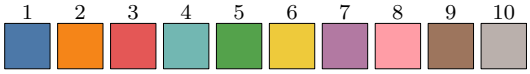
### Spring



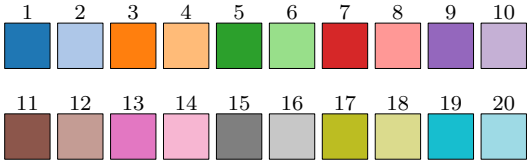

### Summer



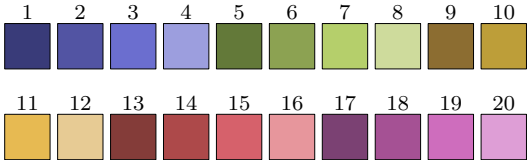

### T10



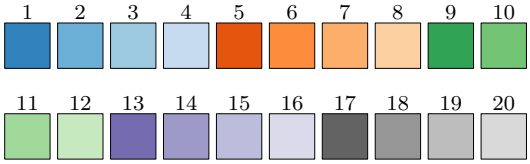

### Tab20



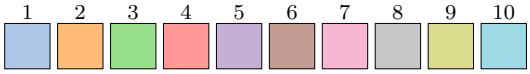

### Tab20b



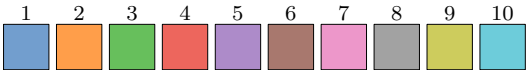

### Tab20c



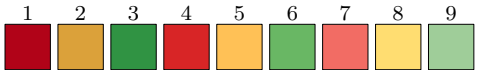

### TableauLight



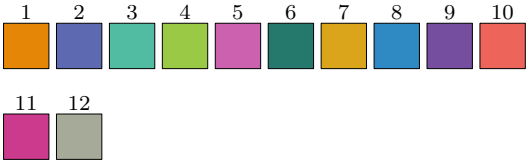
**TableauMedium**



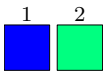

**TrafficLight**



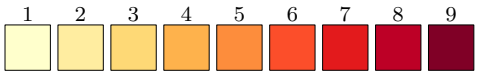

**Vivid**




**Winter**



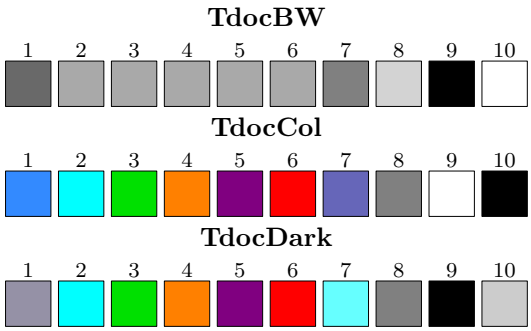

**YlOrRd**




**Zissou**

## Semantic palettes

Semantic palettes assign colors to categories — such as UI states, system status, or pedagogical elements — to convey immediate meaning.



## Sequential palettes

Sequential palettes represent ordered data using a single or multi-hue gradient with monotonic lightness.

**Acton**

**AfmHot**

**AgGrnYl**

**AgSunset**

**Algae**

**Amber**

**Amethyst**

**Amp**

**Apple**

**Arctic**

**ArmyRose**

**Autumn**

**Balance**

**Bam**

**Bamako**

**BamO**

**BatlowK**

**BatlowW**

**Berlin**

**Bilbao**

**Binary**

**Blackbody**

**BlindFish**

**BlueDarkOrange**

**BlueDarkRed12**

**BlueDarkRed18**

**BlueGray**

**BlueGreen**

**BlueGrey**

**BlueOrangeRed**

**BlueRedPLY**

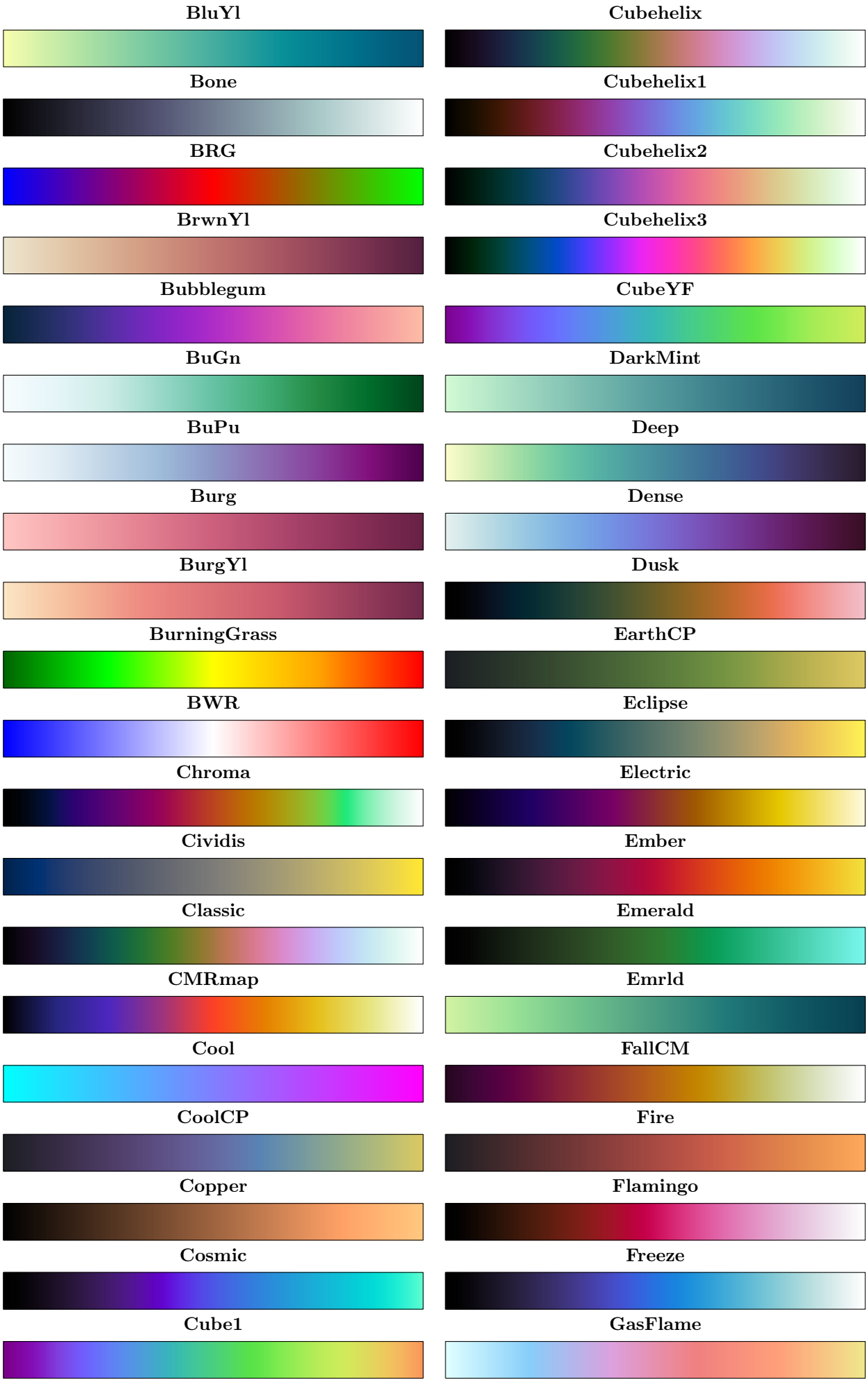
**Blues**

**Blues7**

**Blues10**

**BluesCP**

**BluGrn**


BluYl
Bone
BRG
BrwnYl
Bubblegum
BuGn
BuPu
Burg
BurgYl
BurningGrass
BWR
Chroma
Cividis
Classic
CMRmap
Cool
CoolCP
Copper
Cosmic
Cube1
Cubehelix
Cubehelix1
Cubehelix2
Cubehelix3
CubeYF
DarkMint
Deep
Dense
Dusk
EarthCP
Eclipse
Electric
Ember
Emerald
Emrld
FallCM
Fire
Flamingo
Freeze
GasFlame

**Gem**

**GeoRainbow**

**GhostLight**

**GistEarth**

**GistGray**

**GistHeat**

**GistNcar**

**GistRainbow**

**GistStern**

**GnBu**

**Gnuplot**

**Gnuplot2**

**Gothic**

**Gray**

**Greens**

**GreensCP**

**Greys**

**GreysCP**

**Haline**

**Heart**

**Horizon**

**Hot**

**HotPLY**

**HSV**

**Ice**

**Inferno**

**Jet**

**JetPLY**

**JimSpecial**

**Jungle**

**Lavender**

**Lemon**

**Lilac**

**LinearL**

**Magenta**

**Magma**

**Matter**

**Mint**

**Neon**

**NipySpectral**

**Nuclear**

**Ocean**

**OceanCM**

**Oranges**

**OrangesCP**

**OrRd**

**OrYel**

**Oxy**

**PastelRainbow**

**Peach**

**Pepper**

**PerceptualRainbow**

**Phase**

**Pink**

**Pinks**

**PinkYl**

**Plasma**

**PuBu**

**PuBuGn**

**PuRd**

**Purp**

**Purple**

**Purples**

**PurplesCP**

**PurpOr**

**Rainbow**

**RainbowPLY**

**Rainforest**

**RdPu**

**Red**

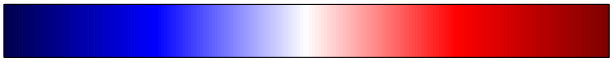
**RedOr**

**Reds**

**RedsCP**

**Sapphire**

**Seismic**



**Sepia**

**ShiftRainbow**

**Solar**

**Speed**

**Spring**

**Summer**

**SunBurst**

**Sunset**

**SunsetDark**

**Swamp**

**Teal**

**TealGrn**

**Teals**

**Tempo**

**Terrain**

**Thermal**

**Torch**

**Toxic**

**Tree**

**Tropical**

**Turbid**

**Turbo**

**Twilight**

**Viridis**

**Voltage**

**Warm**

**Water**

**Wind**

**Yellows**

**YlGn**

**YlGnBu**

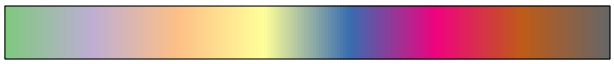
**YlOrBr**

**YlOrRd**

## Stepped palettes

Stepped palettes are designed to highlight distinct shifts rather than smooth progression. They use high-contrast colors that meet at sharp boundary points to emphasize deviation.

**Accent**



**Antique**

**Aquatic1**

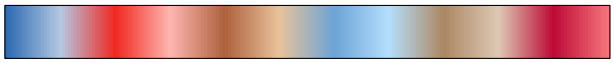
**Aquatic2**

**Aquatic3**

**Batlow**

**BlueOrange**

**BlueRed**



**Bukavu**